Dermatoglyphic Patterns and Academic Performance Among Primary School Pupils in Ekpoma, Nigeria

[*]Anyanwu L.C., Owoyemi B.A., Ishola Z.T.

Department of Anatomy, Faculty of Basic Medical Sciences, College of Medicine, Ambrose Alli University, Ekpoma, Nigeria.

ABSTRACT

Finger/palm dermatoglyphics developed same time as the brain and both have genetic dispensation. Could brain intellectual function leave traces on the fingers/palms? This study, assesses the relationship between dermatoglyphic patterns and academic performance. Three years academic results of Ambrose Alli University Primary School, Ekpoma, were used to sample pupils into those with high or poor academic performance. Finger and palm prints were obtained and analyzed. The data were subjected to descriptive statistics and chi-square test using SPSS (version 21). Pupils with high academic performance were significantly ($P<0.05$) more likely to have whorl fingerprint in the right index ($X^2 = 8.023$), left index ($X^2 = 4.250$) and ring fingers ($X^2 = 4.250$) with females more likely represented in the left index ($X^2 = 9.093$) and ring fingers ($X^2 = 6.982$). According to the number of Primary creases, Intersections of the primary crease, and Complete transverse Crease (PIC) model of classification, the 300 PIC was more likely to occur in the left hand of high academic performing pupils ($X^2 = 4.800$, $P<0.05$) with significant difference ($X^2 = 4.500$; $p<0.05$) in the male. Pupils with high academic performance were more likely to have 3 points origin in the left palm (4.800; $p<0.05$). Academic performance showed a relationship with finger and palm prints and varies with gender. The presence of whorl in the right index left index, and ring fingers, left hand 300PIC, and 3 points origin of primary palm creases maybe determinants for high academic performance.

Key words: Dermatoglyphics, fingerprints, palm prints, intellectual function; academic performance.

[*]**Correspondence: Email:** proflukeanyanwu@gmail.com; **Phone No.:** +2348037081234.

1. INTRODUCTION

Dermatoglyphics is the scientific study of finger and palm epidermal ridges or papillary peak patterns refered to as fingerprints and palmprints[1]. The nature and manner of appearance of the ridges are genetically determined and thus distinct, unique, and permanent[2-4]. Finger and palmprints are not usually influenced significantly after birth by environmental factors[5] for which reason they aid in investigations and identification of individuals[6]. Dermatoglyphics is now seen as a non-invasive tool for identification, and diagnoses of psychogenetic disorders[7] and many studies have correlated finger and palmar prints to psychological characteristics and certain mental disorders[8].

Considering that dermatoglyphic pattern of the fingers and palms as well as the intellect function of the brain are genetically determined or has genetic disposition, one may hypothesize that intellect, learning capability, and memory may present some traces on the finger and/ palm[2, 9]. In this regard, many genetic diseases have been documented to leave traces on the brain and hand with Johnson and Opitz[10] documenting delayed development, learning difficulties, and/or behavioral disorders with a simian or single transverse palmar crease and Sydney crease. Strong correlative power has been reported between simian line and Sydney creases on the palm with mental retardation in Down syndrome, Rubinstein-Taybi-syndrome, Trisomy 18 (Edward's syndrome) and Trisomy 13 (Patau's syndrome) with delayed development, learning difficulties, and/or behavioral disorders in these syndromes[11].

Reports have it that the brain and epidermal ridges of the fingers and palms develop at the same period from embryonic ectoderm[12] and the fingers and palms have high brain innervation density which established the dermatoglyphic characteristics[13]. Could it, therefore, be that the intellect of an individual can be represented in the fingers and palms? Of concern, despite the available studies correlating academic capabilities with dermatoglyphics,no study to the best of our knowledge has been conducted in Nigeria. The aim of this study, is to explore the relationship between finger and palm dermatoglyphic and academic performance.

2. MATERIALS AND METHODS

2.1. Study Area: Ambrose Alli University Primary School, is situated between Ujoelen extension and Alli Square Roadabout in Ekpoma. Ekpoma is the administrative head quarters of Esan West Local Government Area, Edo State, Nigeria. The Local Government Area is located approximately on longitude 8°5' East of the Greenwich meridian and latitude 7°23' North of the equator. Ekpoma occupies an area of 502km^2 and has a population of over 125,842[26]. Ekpoma is host to several public and private institutions including the state owned University, Ambrose Alli University. It is also a home to multi-ethnic nationalities predominantly the Esan, Benin, Igbo, Hausas, Yoruba and people from other neighbouring communities. The inhabitants are mostly farmers, traders, public servants and students.

2.2. Sample population: The subjects for the study were 40 primary school pupils of the Ambrose Alli University, Ekpoma. Purposive sampling approach was applied in the selection of 5 each of best and poor performing pupils from classes 3 to 6. Ambrose Alli University Primary School is a mixed school with pupils population of over 900. The male to female ratio is 1.2:1.

2.3. Inclusion criteria: Participants included male and female primary school pupils between the ages of 6 and 13.

2.4. Exclusion criteria Participants who gave consent and had any distortions of fingers and palm or infected hand as well as wounds on fingers and palms prompting scars were excluded.

2.5. Ethical consideration: Before the study, an informed consent statement was fully explained and written consent was obtained from the management of Ambrose Alli University Primary School, Ekpoma. Head of classes were then informed by the management and their consent was also obtained. Participants'parents or guidance were now properly informed about the objectives, relevance, and purpose of the study in writing, and they were assured about the confidentiality of the data.

2.6. Academic performance: The average scores for three consecutive term examinations were calculated for each student and stratified into high (70% - 100%) or low (less than 35%). The relationship between a student's academic score and dermatoglyphic patterns was then explored using relevant statistics. 20 each of high and poor academic performance pupils were recruited from class 3 to 6 into the study after evaluating their report cards for three consecutive terms.

2.7. Sample collection: Fingerprint patterns were studied on the 10 fingers using a stamp ink and white A4 sheets as documented in Offei *et al*.[1] with some modification. Briefly, the stamp ink was applied to the pulp area of each finger, which was then pressed on the white A4 sheets. Each finger was done one after another and labeled accordingly due to the opposing orientation of the thumb. Numbering of the digit was done from thumb to little finger.

Right and left palm pictures were obtained using a mobile camera with 20mp resolution. Briefly, participant's palms were clean, dried,

and gently placed side by side for clear palm prints presenting all the ridges and triradiate. Pictures were printed on A4 plain papers and for purpose of confidentiality, identifying codes were used to keep participants' privacy, and the age and sex of the participants and sides of the palms were recorded.

2.8. Sample analysis: Fingerprints were studied with the help of a magnifying lens by grouping into the three basic pattern types as distinguished by Galton[14] to be whorl, loop, and arch. Arch pattern ridges enter from one side, traverse a pattern area, from a curve and leave from the other side. Loop ridges enter on one side of the digit, recurve abruptly and leave the pattern area on the same side. The Whorl is the complex pattern defined as any ridge configuration with two or more triradii (see figure 1).

Palm prints were analyzed using the number of Primary creases, Intersection of primary creases, and Complete transverse crease (PIC) criteria as in Mekbeb[15] as in figure 2. Abnormal creases were identified based on the relationships between the PTC and DTC, as Simian, Sydney, and Suwon creases (see figure 3). According to the manner of origins of the three primary creases on the radial side of the palm, creases were classified into one, two, or three points of origin[16, 17]. If both palms have the same PIC profile, it was termed symmetry otherwise it was labeled asymmetry. Furthermore, each of the three major creases was studied in detail for any variants and was classified according to Park *et al*.[18] as branched, forked, accessory, broken, or cascade. Identifications of palmar creases were done on both right and left hands (see figure 4)

2.9. Statistics: The obtained data were entered into MS Excel and analyzed using statistical tools from Statistical Package for Social Sciences (SPSS) version 21.0. The results were compared between high and low academic performance pupils using Chi-Square and Student's t-test analysis where applicable. The critical value for statistical significance was set at 0.05.

3. RESULTS

The studied pupil comprises of 20 each of high and poor academic performance determined from their three years previous results. They were sampled from primary 3 to 6 and five pupils each were sampled from each class for the study. The mean ages of the pupils with high and poor academic performance were 9.60±1.47 years and 9.50±1.50 years respectively. The high-performance group comprises 12 girls and 8 boys while the poor performance was 10 each of boys and girls.

Table 3.1 is a comparative table of the various fingerprints patterns between high and poor academic performance pupils in the right and left hands. Fingerprint patterns show significant differences with academic performance in the right index, left index, and ring fingers. Specifically, pupils with high academic performance are more likely to present with whorl in their right ($X^2 = 8.023$; $P<0.05$) and left ($X^2 = 4.250$; $P<0.05$) index fingers as well as the left ring finger ($X^2 = 4.250$; $P<0.05$). On the other hand, the pupils with poor academic performance were more likely to present with loop fingerprints in both fingers of the palm.

Tables 3.2 and 3.3 present right and left hands comparative table of gender differences in the various fingerprints patterns between high and poor academic performance pupils respectively. There was no significant difference ($p>0.05$) in the right hand with the various fingerprints patterns with academic performance between gender. On the left hand, however, females with high academic performance are more likely to present with whorl in the index ($X^2 = 9.093$; $p<0.05$) and ring ($X^2 = 6.982$; $p<0.05$) fingers compared with the female with poor academic performance.

Table 3.4 presents a comparative table of the palm prints patterns according to the PIC profile between high and poor academic performance pupils in the right and left palm. The predominant occurring PIC profiles in the right hand were 300 and 310 PIC in both pupils with high and poor academic performance. The 200 and 311 PIC profiles were absent in the right hand of pupils with poor academic performance. Chi-square analysis revealed no relationship between right-hand PIC profile with academic performance. The predominantly occurring PIC profile in the left hand was the 310 PIC for both pupils with high and poor academic performance; however, the 300 PIC is more likely to occur in the left hand of pupils with high academic performance. Chi-square analysis revealed a statistically significant and the relationship between left-hand 300 PIC profile with high academic performance (X^2 = 4.800, p = 0.028; P<0.05).

Figure 3.1 represents a bar chart of the distribution of PIC symmetry in both palms amongst high and poor academic performance pupils. Chi-square analysis revealed no relationship in PIC symmetry in both palms with academic performance.Table 5 presents the gender differences in palm prints PIC patterns between high and poor academic performaning pupils. There was no significant difference (p>0.05) between academic performance and the various palm prints PIC patterns in the right hand of both male and female as well as in the left hand of female pupils.

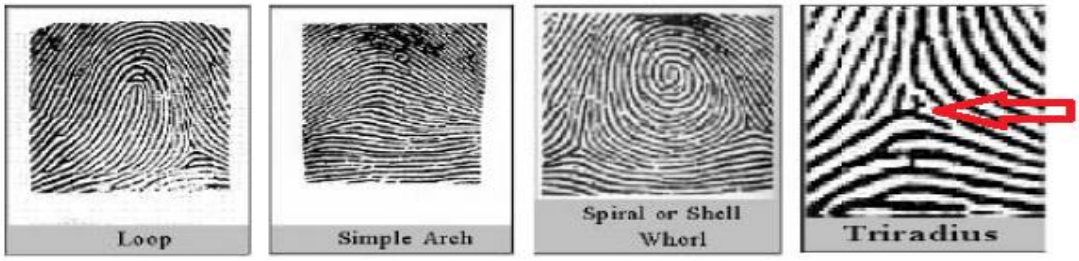


Figure 1. Patterns of fingertips

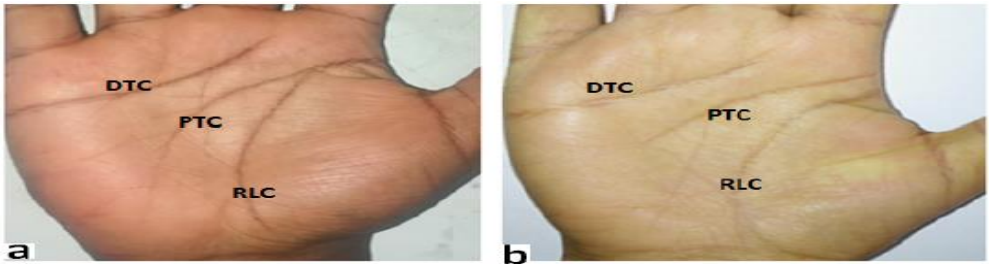


Figure 2. PIC determination of the palms

Where DTC: distal longitudinal crease; PTC: proximal longitudinal crease; RLC: radial longitudinal crease



Figure 3. Abnormal palmar creases

Where (a) Simian crease, PTC and DTC have fused and spans the entire width of the palm; (b)Sydney crease, PTC crosses the entire width of the palm to near the ulnar border; and (c) Suwon crease, an extended DTC that spans the entire width of the palm.

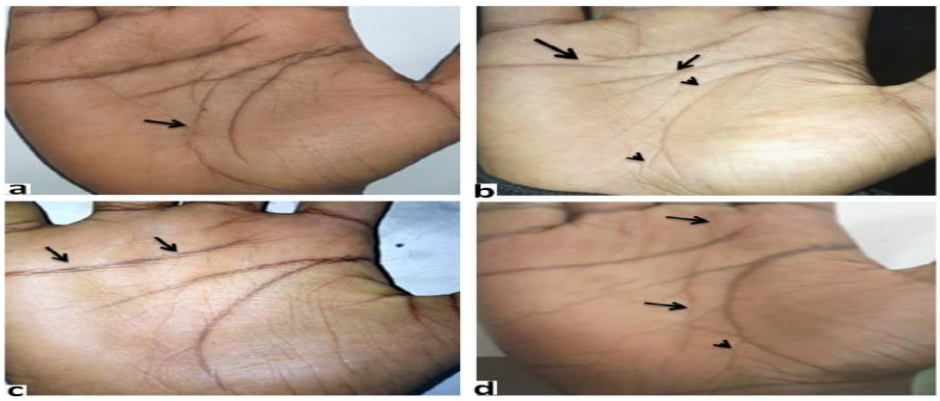


Figure 4. Variants of palm creases

Arrow in "a" shows the accessory crease of the RLC. In "b" big and small arrows show, respectively broken DTC and forked PTC, while arrowheads show branches of RLC; c: the cascade of DTC (arrows); d: middle finger crease (arrows) and branched RLC (arrowhead).

Table 3.1. Comparative table of the various fingerprints patterns between high and poor academic performaning pupils

Hand	Prints	Finger Thumb		Index		Middle		Ring		Little	
		High	Poor	High	Poor	High	Poor	High	Poor	High	Poor
	Arch	3	3	4	5	2	5	2	3	4	2
	Loop	6	12	6	13	13	13	12	12	12	17
Right	Whorl	11	5	10	2	5	2	6	5	4	1
	X^2; p-value	4.250 p>0.05		8.023 P<0.05		2.571 p>0.05		0.291 p>0.05		3.329 p>0.05	
	Arch	3	6	6	9	6	5	2	1	5	3
	Loop	7	10	3	8	8	13	8	16	13	16
Left	Whorl	10	4	11	3	6	2	10	3	2	1
	X^2; p-value	4.101 p>0.05		4.250 P<0.05		3.281 p>0.05		4.250 P<0.05		4.250 p>0.05	

Table 3.2.Right-hand comparative table of gender differences in the various fingerprints patterns between high and poor academic performaning pupils

		High performance			Poor performance				p-value
Finger	Gender	Arch	Loop	Whorl	Arch	Loop	Whorl	X^2	
Thumb	M	0	3	5	1	8	1	5.789	0.055
	F	3	3	6	2	4	4	0.566	0.754
Index	M	1	3	4	2	7	1	3.555	0.169
	F	3	3	6	3	6	1	4.426	0.109
Middle	M	0	6	2	2	7	1	2.215	0.330
	F	2	7	3	3	6	1	1.104	0.576
Ring	M	0	7	1	1	6	3	1.878	0.391
	F	2	5	5	2	6	2	1.205	0.548
Little	M	1	6	1	0	10	0	2.812	0.245
	F	3	6	3	2	7	1	1.104	0.576

Where M = male and F = female

Table 3.3. Left-hand comparative table of gender differences in the various fingerprints patterns between high and poor academic performaning pupils

		High performance			Poor performance				p-value
Finger	Gender	Arch	Loop	Whorl	Arch	Loop	Whorl	X^2	
Thumb	M	0	4	4	2	6	2	2.880	0.237
	F	3	3	6	4	4	2	2.121	0.346
Index	M	2	2	4	3	4	3	0.797	0.671
	F	4	1	7	6	4	0	9.093	0.011
Middle	M	1	5	2	3	6	1	1.217	0.544
	F	5	3	4	2	7	1	4.541	0.103
Ring	M	0	5	3	0	8	2	0.678	0.410
	F	2	3	7	1	8	1	6.982	0.030
Little	M	0	7	1	0	10	0	1.324	0.250
	F	5	6	1	3	6	1	0.321	0.852

Where M = male and F = female

However, males with high academic performance were more likely to present with the 300 PIC pattern in the left hand (X^2 = 4.500; p = 0.034; p<0.05).

Table 3.6 presents a comparative table on points of origin of palm prints PIC in the right and left palms between high and poor academic performaning pupils. There was no relationship in points of origin of palm prints PIC in the right palm with academic performance. However, pupils with high academic performance are more likely to present 3 points of origin in the left palm and Chi-square analysis revealed a statistical relationship (4.800; p = 0.028, p<0.05). Figure 6 represents a bar of the distribution of point of origin of palm prints symmetry in both palms amongst high and poor academic performaning pupils. Although pupils with poor academic performance were more likely to present symmetrical points of origin, Chi-square analysis revealed no statistical relationship

Table 3.4. Comparative table of the palm prints patterns according to PIC between high and poor academic performaning pupils

Hand	PIC	High performance	Poor performance	X^2	p-valve
Right	200	2	0	4.800	0.187
	300	8	8		P>0.05
	310	8	12		
	311	2	0		
Left	300	8	2	4.800	0.028
	310	12	18		P<0.05

Table 3.7 presents gender differences in palm prints point of origin between high and poor academic performaning pupils. Palm print's points of origin showed no relationship between gender and academic performance in the right hand of both male and female pupils and the left hand of female pupils. However, Chi-square analysis revealed a statistical relationship (X^2 = 4.500; p = 0.034; p<0.05) between male left-hand points of origin and academic performance as males with high academic performance were more likely to present with 3 points of origins in the left palm.

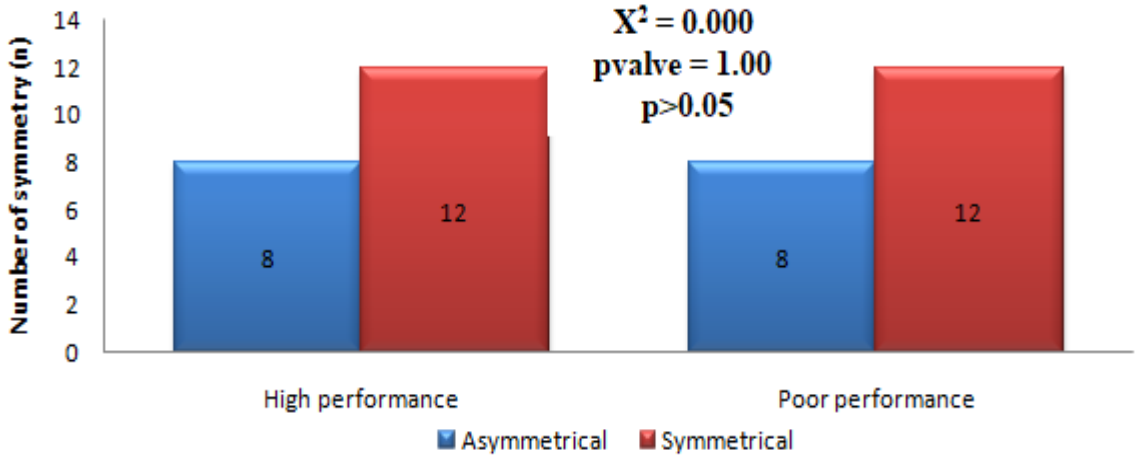


Figure 5.A bar chart showing the distribution of PIC symmetry amongst high and poor academic performaning pupils.

Table 3.5. Comparative table of gender differences in palm prints PIC patterns between high and poor academic performaning pupils

		Right hand				Left hand			
Gender	**PIC**	**High**	**Poor**	**X^2**	**p-valve**	**High**	**Poor**	**X^2**	**p-valve**
Male	**300**	2	5	3.214	0.200	3	0	4.500	0.034
	310	4	5		p>0.05	5	10		P<0.05
	311	2	0			0	0		
Female	**200**	2	0	3.667	0.160	0	0	1.180	0.277
	300	6	3		p>0.05	5	2		p>0.05
	310	4	7			7	8		

Table 3.6. Comparative table on points of origin of palm prints between high and poor academic performaning pupils

Hand	Points of origin	High performance	Poor performance	X^2	p-valve
Right	2	12	14	0.440	0.507
	3	8	6		p>0.05
Left	2	12	18	4.800	0.028
	3	8	2		P<0.05

Table 3.8 compares the various primary palm prints variants with academic performance. Although Chi-square analysis revealed no statistical relationship between academic performance and primary palm prints variants, variants in the primary palm prints are more likely to occur in pupils with high academic performance except in the right hand PLC and left hand RTC.

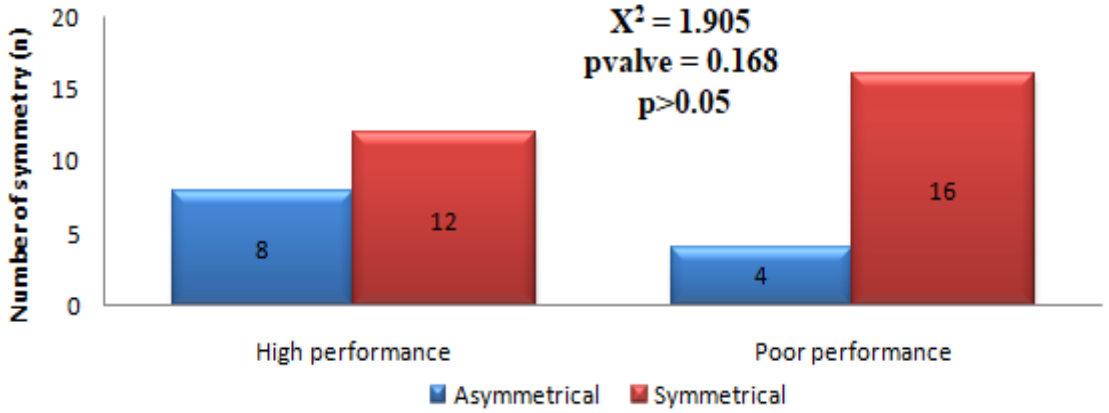


Figure 6. A bar chart showing the distribution of point of origin of palm prints symmetry amongst high and poor academic performaning pupils.

Table 3.7.Comparative table of gender differences in palm prints point of origin between high and poor academic performaning pupils

		Right hand				Left hand			
Gender	**Point of origin**	**High**	**Poor**	**X^2**	**p-valve**	**High**	**Poor**	**X^2**	**p-valve**
Male	**2**	6	7	0.055	0.814	5	10	4.500	0.034
	3	2	3		p>0.05	3	0		P<0.05
Female	**2**	6	7	0.903	0.342	7	8	1.180	0.277
	3	6	3		p>0.05	5	2		p>0.05

Table 3.8. Comparative table on the various primary palm prints variants between high and poor academic performaning pupils

		Right hand				Left hand			
Crease	PIC	High	Poor	X^2	p-valve	High	Poor	X^2	p-valve
DLC	Non-variant	7	9			7	10		
	Broken	3	2			5	3		
	Branched	2	1	0.783	0.941	2	2	1.229	0.873
	Cascade	3	3		p>0.05	3	2		p>0.05
	Forked	5	5			3	3		
PLC	Non-variant	5	3			4	7		
	Accessory	0	0			1	1		
	Broken	2	2	2.500	0.645	0	2	5.334	0.376
	Branched	12	12		p>0.05	12	7		p>0.05
	Cascade	0	2			0	1		
	Forked	1	1			3	2		
RTC	Non-variant	4	6			3	1		
	Accessory	4	7			7	2		
	Broken	2	1	4.618	0.464	1	6	9.683	0.085
	Branched	3	2		p>0.05	1	3		p>0.05
	Cascade	2	3			1	3		
	Forked	5	1			7	5		

4. DISCUSSION

Studies investigating the correlation of finger and palm dermatoglyphics analysis with other phenotypic features and diseases are increasing lately. Analysis of the fingerprint patterns from this study showed the whorl and loop patterns were the most commonly occurring pattern in pupils with high and poor academic performance respectively. However, primary school pupils with high academic performance were significantly more likely to have whorl in the right-hand index finger, left-hand index, and ring fingers compared with those of poor academic performance with loop prevalence in the fingers of both hands.

These findings suggest an association between academic performance and fingerprint patterns. This suggestion agrees with the report by Offei *et al*.[1] who demonstrated a striking relationship between dermatoglyphic pattern and student learning capability in a normal population of students in a Ghanaian Secondary School.

In line with the observed presence of the whorl fingerprint pattern in pupils with high academic performance, Parker[19] had reported a statistically significant amount of whorl in individuals with higher than normal (I.Q 110-129) and normal (I.Q 90-109) intellectual ranges compared to individuals with below-normal intellectual ranges (70-89). In a study among male Vietnamese students, higher IQ was shown to correlates with the presence of whorls on the right thumb, forefingers, and loops on left forefingers and in female Thai students, whorls on left thumb, right thumb, and loop on left forefingers and right forefingers[20]. Also, the study by Offei *et al*.[1] reported the central pocket loop whorl on any finger relates significantly to better academic performances of students ($p<0.0001$) compared to the other patterns. However, Nayak *et al*[21] reported the whorl pattern correlates with low or average scoring groups as against the ulnar loop pattern among higher scores and better learning methodologies among undergraduate medical students belonging to three different Malaysian ethnic races.Another study among adolescents has also documented the loop pattern of fingerprints on any finger correlates significantly with better IQ[22].This contradicting report by Nayak *et al*.[21] and Najafi[22] indicates that the association of academic performance with dermatoglyphic patterns may vary with geographical distribution.

The most prevalent palm print patterns observed in both hands in this study were the 300PIC and 310PIC. Interestingly, the 200PIC and 311PIC were absent in the right hand of pupils with poor academic performance. There was no significant relationship in the different PIC patterns in the right hand. However, in the left hand, the pupils with high academic performance were more likely to present the 300PIC as against the 310PIC among pupils with poor academic performance and there was a statistically significant relationship ($X^2 = 4.800$; $p = 0.028$; $p<0.05$) which was also observed to be significant in the male ($X^2 = 4.500$; $p = 0.034$; $p<0.05$). The dominance of 300PIC and 310PIC in this study agrees with the study in China population by Wu *et al*.[23] and Fang *et al*.[24] and in Ghanaian population in a study by Offei *et al*.[21]. In agreement with the present findings, the study by Ba-Etilayoo and Osabutey[25] showed a statistically significant ($p < 0.05$) association between academic status and PIC patterns on the left hand and reported participants with PIC 300, 310, and 311 were more likely to be lower academic achievers.

There was no significant relationship or difference between the academic performances of students with symmetrical palm prints and those with asymmetrical palm prints in the present study. However, the study by Offei *et al*.[1] showed students with symmetrical palm prints are three times likely to obtain high scores than their asymmetrical

counterparts. Offer *et al*.[1] also concluded that functional asymmetry is not absolute as the degree may be associated with the individual's inherent ability. Although there was no significant difference in the primary crease variant, pupils with high academic performance were more likely to present DLC and RTC variants as compared to the higher PLC variant in pupils with poor academic performance.

5. CONCLUSION

The study has demonstrated an association between the dermatoglyphic patterns observed on the fingers and palms and student's academic performance with gender also contributing to the correlation. From findings, students with the whorl fingerprints in the right index left index, and ring fingers performed better than those with other finger prints. Also, the presence of the 300PIC especially in the left hand is a determinant of high academic performance. The results of the present study could serve as an accessory method that can be applied for educational performance evaluation, screening, and helping pupils with difficulty in learning. For this reason, there is a need to explore the dermatoglyphic patterns of individuals for the determination of academic achievement and attainment.

ACKNOWLEDGEMENTS

Our appreciation goes to the study participants, without whom this study would have been impossible. We also appreciate their parents and the management of Ambrose Alli University Primary School, Ekpoma. We also extend our thanks to Mr.AkpamuUwaifoh for the assistance he provided in statistical analysis.

COMPETING INTERESTS

The authors declare that they have no competing interests.

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